

Evgeniia Tokarchuk

Research Scientist in NLP

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SUMMARY

Research Scientist in NLP focused on representation learning and geometry-aware optimization of Transformer models. 6+ years' experience, first-author at EMNLP, TMLR, and NAACL, with demonstrated gains in robustness, efficiency, and retrieval. Hands-on with PyTorch, FAISS, and HPC-scale systems.

SKILLS

- **Machine Learning & Research:** representation learning, transformer models, embeddings geometry, optimization, evaluation, data pipeline, retrieval methods
- **Technical Stack:** python (10+yrs), pyTorch, fairseq, linux, bash shell, git, slurm, faiss
- **Languages:** english (C1), russian (native), dutch (A2)

WORK EXPERIENCE

University of Amsterdam, Amsterdam, The Netherlands

- Research Assistant (Machine Learning & NLP), Language Technology Lab May 2021 – present
 - Led end-to-end **NLP research** (problem → publication), delivering 4+ first-author papers (EMNLP, TMLR, NAACL, EACL)
 - Designed and ran **large-scale experiments** using **PyTorch/Fairseq** on **Slurm HPC**
 - Introduced geometry-aware **optimization** improving **retrieval** quality and decoding **efficiency**, and presented results at international venues.

Stichting Volunteers with no limits, Amsterdam, The Netherlands

- Chairwoman Mar 2023 – present
 - Lead strategic and operational coordination of a volunteer-run charity supporting Ukraine, overseeing donation campaigns, partnerships, and distribution efforts
- Logistic Coordinator Mar 2022 – Mar 2023
 - Managed collection, sorting, and distribution of humanitarian aid; coordinated volunteers and supply flows to ensure timely delivery of essential goods

eBay Aachen, Aachen, Germany

- Working Student (Machine Learning & NLP) Oct 2019 – Mar 2021
 - Designed and implemented **end-to-end training** for pivot-based neural machine translation using non-autoregressive **Transformer** models.
 - Applied **reinforcement learning** techniques to optimize sequence-level objectives in pivot-based translation systems.
 - Collaborated with applied research engineers on production-oriented ML workflows and evaluation.

Machine Learning and Human Language Technology Lab, RWTH Aachen, Germany

- HiWi (Machine Learning & NLP) Feb 2018 – Sep 2019
 - Built and optimized NER data and training pipelines in Flair (**PyTorch**) and conducted large-scale character- and subword-level NMT experiments with RETURNN (TensorFlow) on **HPC infrastructure**.

PressFoto R&D department, Chelyabinsk, Russia

- Algorithm's Developer Oct 2016 – Aug 2017
 - Worked on embedding-based similarity search to improve image **retrieval** quality for text-driven queries.
 - Optimized ranking and matching algorithms, focusing on relevance and retrieval precision in large-scale media datasets.

The Honeynet Project,

- Google Summer of Code 2016, May 2016 – Aug 2016
 - Designed and implemented SNARE and TANNER honeypot systems from scratch using Python and aiohttp.
 - Built scalable, asynchronous services for collecting and analyzing malicious network activity.
 - Delivered production-quality open-source contributions adopted by the security research community.

EDUCATION

University of Amsterdam, Amsterdam, The Netherlands

- PhD Candidate, Language Technology Lab May 2021 – present
 - My research under supervision of Dr. Vlad Niculae has been focused on representation collapse in Transformer-based models and how it can be countered with geometry-aware optimization.

RWTH Aachen University, Aachen, Germany

- **M.Sc. Software Systems Engineering** Oct 2017 – Mar 2021
 - GPA: 1.7 (1.0 best; 5.0 worst)
 - **Relevant Coursework:** Statistical Classification and Machine Learning, Natural Language Processing, Machine Learning, Advanced Machine Learning, Automatic Speech Recognition (Lab)
 - **Master thesis topic:** Tight model integration for pivot-based Neural Machine Translation

South Ural State University, Chelyabinsk, Russia

- **Specialist in Information Security for Automated Systems** Sep 2011 – Jun 2016
 - Graduated with Honors

- SELECTED PAPERS**
- **E. Tokarchuk**, M.K. Nachesa, S. Troshin, V. Niculae. Representation Collapse in Machine Translation Through the Lens of Angular Dispersion, *EACL*, March 2026 (to appear) → Analysis on the dynamics of representation collapse in Transformer.
 - **E. Tokarchuk**, S. Troshin, V. Niculae. Angular Dispersion Accelerates k -Nearest Neighbors Machine Translation. *EMNLP*, November 2025
 - **E. Tokarchuk**, H.C. Bakker, V. Niculae. Keep your distance: learning dispersed embeddings on $\mathbb{S}_m$. *TMLR*, August 2025.
 - **E. Tokarchuk**, V. Niculae. The Unreasonable Effectiveness of Random Target Embeddings for Continuous-Output Neural Machine Translation. *Meeting of the North American Chapter of the Association for Computational Linguistics (NAACL)*. June 2024 (oral presentation).
 - **E. Tokarchuk**, J. Rosendahl, W. Wang, P. Petrushkov, T. Lancewicki, S. Khadivi and H. Ney. Towards Reinforcement Learning for Pivot-based Neural Machine Translation with Non-autoregressive Transformer, *Reinforcement Learning for Real Life Workshop*, July 2021.

**ACADEMIC
SERVICE &
MENTORSHIP**

- **Teaching Assistant, NLP 2**, University of Amsterdam Apr 2022 – Jun 2024
 - Supervised projects on reinforcement learning and MBR for non-autoregressive NMT.
- **IAI community mentor**, University of Amsterdam Nov 2023 – present
 - Inclusive AI (IAI) community for the Master’s programme AI mentor
- **M.Sc and B.Sc. thesis supervisor**, University of Amsterdam Nov 2021 – Aug 2024
 - 3 B.Sc and 2 M.Sc. thesises
- **Conference reviewer** 2021 – present
 - ACL*, NeurIPS, ICML, ICLR

**AWARDS &
SCHOLARSHIPS**

- **DAAD Study Scholarships for Graduates of All Disciplines** 2018 – 2019
- **South Ural State University Graduate Award** 2016
 - For excellent academic performance, diligence and perseverance in mastering the program of study, scientific curiosity and creative approach to research work.